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Research Paper

Numerical study on photovoltaic thermal phase change material system in hot climatic conditions

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ABSTRACT

High operating temperature of photovoltaic (PV) panels hinder electrical and thermal efficiency, leading to a decrease in performance. To address this issue, researchers have explored various cooling techniques. Phase change materials (PCMs) can be seen as a promising solution to this problem However, the limitations of the PV/PCM system is that, the continuous increase in PCM temperature after melting and the inability to use the heat stored in the PCM. These limitations can be overcome by using a simplified passive water-based PVT/PCM system. In the present study, two separate containers, one filled with PCM and the other with water are used. They are integrated with a PV panel on the backside. Initially, a constant PCM thickness of 50 mm and water container thicknesses (WCT) of 20 mm, 30 mm, and 40 mm were considered to determine a suitable WCT for the proposed PVT/PCM system and the performance was investigated at 90° system orientation. A 2-D computational model is developed and analyzed using ANSYS FLUENT software to determine suitable WCT with PV/PCM system at different system orientations. The preliminary investigation shows that a WCT of 30 mm is the best arrangement for the proposed PVT/PCM system. As the WCT is increased beyond 30 mm, no significant change has been observed in PCM melt fraction and PV panel temperature. Further investigation shows that an increase in the system orientation from 30° to 90° decreases the average temperature of the photovoltaic (PV) panel and phase change material (PCM). The complete melting of PCM results in an average decrease in PV panel temperature of 5.88% (23 °C) and 1.36% (5 °C) at 30° and 90° system orientation, respectively. This decrease in temperature resulted in an overall increase in average electrical efficiency of 14.93% and 1.35% at 30° and 90° system orientations, respectively.

1. Introduction

The growth of any country is evaluated based on the energy potential available for various sectors. Nowadays, many countries are striving for clean and renewable forms of energy. After wind energy, solar energy is one of the most prominent contributing energy sources. The future of solar energy deliberates on technologies for converting solar energy into electricity, i.e., solar photovoltaic (PV) and concentrated solar power (CSP). The energy technologies in these classes will dominate solar power generation between now and 2050 [1]. Solar photovoltaic technology is a rapidly developing field that can be applied to both building integration and remote areas [2]. It saves the environment by decreasing carbon emissions. A major disadvantage of solar photovoltaics is that when the temperature of the PV cells exceeds 25 °C, the electrical efficiency starts to decrease at a rate of 0.4–0.65 %/°C [3].

The techniques used for cooling of PV panels are categorized as active and passive methods. Active methods use external power sources to remove heat, while passive methods rely on natural circulation [4,5]. Active cooling methods can effectively reduce the temperature of PV panels [6], but they come with high costs of installation, maintenance and high power consumption [7-10]. On the other hand, passive cooling methods are not as effective, particularly during peak solar hours, due to their low heat extraction rates and dependence on ambient conditions [11]. These limitations have prompted researchers to investigate new solutions that can efficiently work during peak sunshine hours and maintain stability over a long period of time [12]. The focus of research is on developing efficient cooling methods to improve the electrical and thermal performance of PV systems in hot climatic conditions. The advancement in cooling technology for photovoltaic systems in hot climates to improve both electrical and thermal efficiency is illustrated in Fig. 1.

The use of phase change materials in photovoltaic cooling has recently gained attention as an alternative passive solution [11]. These

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Nomenclature

A_{mush}	mush zone constant
C	specific heat (J/kg-K)
g	acceleration of gravity (m^2/s)
h_o	convection coefficient (w/m^2-K)
h_{ref}	reference enthalpy (J/kg)
I_T	solar insolation (w/m^2)
k	thermal conductivity (w/m-K)
P	pressure (N/m^2)
Q_{th}	thermal energy (kJ)
S	source term
T	temperature (K)
T_p	photovoltaic panel temperature
T_w	water temperature
u	velocity component (m/s)
GNP	gold nanoparticles
PCM	phase change material
PEG	polyethylene glycol
PV	Photovoltaic
PVT-PCM	photovoltaic thermal phase change material
WCT	water container thickness

Greek letters

Υ	melt fraction
β	thermal expansion coefficient
ρ	density (kg/m^3)
μ	dynamic viscosity (kg/m-s)

Subscripts

D	dimensional
ι	liquid
p	pressure/panel
s	solid
th	thermal
w	water

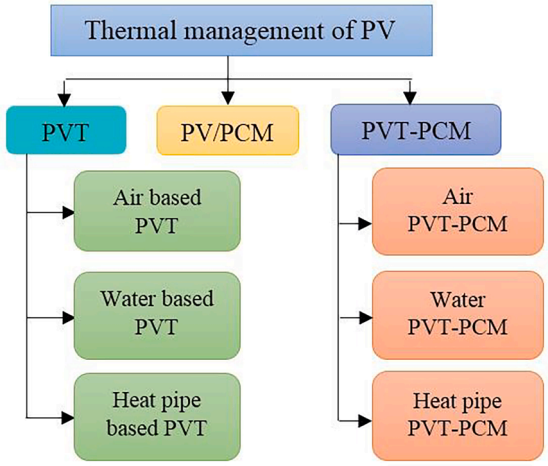


Fig. 1. Development of cooling systems for thermal management of PV.

materials possess the ability to effectively extract heat without the external energy source and require minimal maintenance. A typical PV/PCM system is composed of a photovoltaic panel on top, and an aluminum container filled with PCM is attached on the backside [13]. Many researchers are carrying out experimental and numerical studies, to optimize PV/PCM cooling systems. Nižetić et al. [14] compared the use of several smaller PCM-filled enclosures to one solid PCM-filled rectangular enclosure in PV systems and found that the smaller enclosures led to a performance increase of 10.7%. Also, compared to a full PCM container, the cost of the PCM materials and aluminum used in the smaller enclosures were found to be 47% and 36% less, respectively. Akshayveer et al. [15] introduced the idea of using bifacial PV/PCM systems to improve thermal and electrical performance. It is found that the use of reflecting mirrors with these systems increases the utilization of incident solar radiation by 87% and results in a 77% improvement in power output compared to traditional PV/PCM systems. Foteinis et al. [16] studied the effects of the use of PCM cooling in PV systems in the Mediterranean climate over a long-term period. The result shows that the use of PCM cooling leads to a 9.4% increase in photovoltaic energy output and a high energy payback period of 14 years, compared to a 5 year payback period for traditional PV systems. According to life cycle assessment, the use of PCMs for cooling can also increase the environmental impact by 21.9%. Jaharomi et al. [17] evaluated the performance of a PV system by incorporating nanostructured materials (GNP and CuO) into PCM (PEG 1500) to improve thermal conductivity. The study found that the use of a mixture of GNP-CuO (3% by weight) leads to a 91.81% increase in thermal conductivity, which results in a 6.6 °C drop in temperature of PV panel and a 3% increase in electricity generation. Velmurgun et al. [18] studied the use of a radiation-based phase change material matrix to cool PV systems in Thailand's environmental conditions. The PCM-filled container with a 6 mm gap from the backside of the PV panel, decreases the maximum panel temperature by 13.20 °C. Al-Najjar et al. [19] proposed a new mathematical model for simulating PV/PCM systems as an alternative to using the commercial software ANSYS FLUENT and COMSOL. The proposed model was implemented in MATLAB, using simple numerical operations rather than mesh patterns, making it easier to execute. Xu et al. [20] examined the use of PCM as a cooling medium for PV systems and evaluated the impact of the inverse phase transition from molten to solidified PCM on the temperature of PV panels after cooling. The results show that the average electrical efficiency and power output increase by 1.63% and 1.35 W, respectively within 300 min. It is also reported that PCM-cooled panels took a maximum of 480 min to cool down to room temperature, while non-PCM cooled panels cooled down to room temperature in 80 min.

When integrating PCM with photovoltaics, the thermal properties of the PCM such as thermal stability, low thermal conductivity, supercooling, and volume change during phase transition, can limit the effectiveness of using PCM for cooling PV systems [21]. However, the thermal performance of the system can be improved by using heat transfer enhancers (HTE) such as fins, microencapsulation, metal foams, matrix, nanoparticles, etc. [22,23]. These enhancers can improve the heat transfer rate and increase the thermal efficiency of the system.

During experiments, the system orientation of the location is one of the most crucial factors that affect the performance of PV panels. The solar radiation that impinges on the surface of PV panels depends on the system orientation, thus it affects the melting behavior of PCM and the overall performance of the system. Therefore, the system orientation is important to be considered in the design of PV/PCM systems. Sathe et al. [24] investigated the impact of the angle at which the system is inclined and the thickness of the PCM on the time required for the PCM to melt. The analysis was carried out at system orientations of 30°, 45°, 60°, and 90° and PCM thicknesses of 20 mm, 30 mm, and 40 mm. The results show that the 90° system orientation requires the least amount of time for complete PCM melting in comparison to other orientations. It was also found that as the system orientation decreases, the time required for

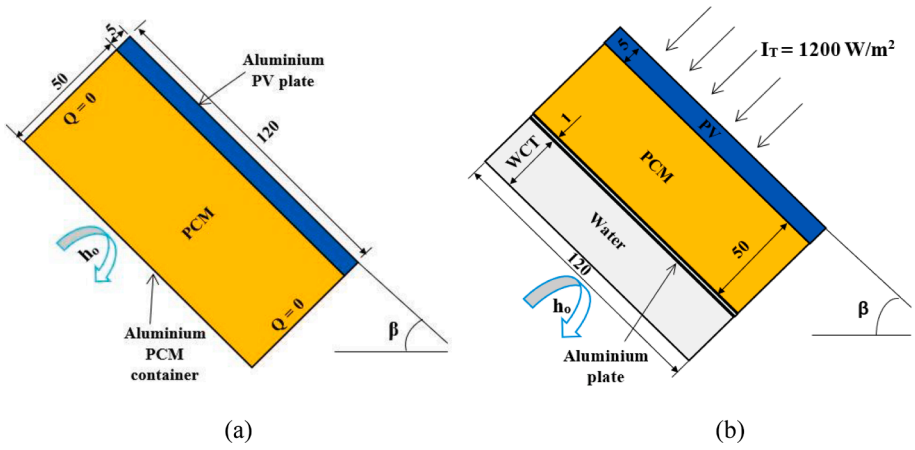


Fig. 2. Schematic illustration of the (a) Reference PV/PCM system and (b) proposed PVT/PCM system.

melting increases. The time needed for complete PCM melting of a 40 mm PCM thickness at a system orientation of 90° is 160 min, while at a system orientation of 30° it is 300 min. These results indicate that both the PCM thickness and system orientation can have a significant impact on the performance of PCM-based thermal management systems. A similar investigation carried out by Abdulmunem et al. [25] found that as the system orientation of a photovoltaic (PV) system increases from 0° to 90°, the temperature of the PV cells decreases as the convective heat transfer increases. The study also suggests that adjusting the system orientation can be an effective way to control the temperature of PV cells and potentially improve the performance of the system.

Incorporating a phase change material (PCM) into a photovoltaic thermal (PVT) system, the overall performance and stability of the system can be improved and maintained over a long period of time. The combination of the PV and thermal components, along with the PCM, enhances the system's efficiency and stability, providing a more reliable energy solution [12]. Su et al. [26] examined the effect of PCM layers at various positions in air-based PVT systems. The result shows that the overall efficiency of a photovoltaic-thermal collector in the upper phase change material mode was 10.7% higher than the no phase change material mode. Moreover, a 3 cm thick layer of phase change material in the upper mode of a photovoltaic-thermal collector was verified to provide excellent results in terms of both electrical and thermal performance. Shastry et al. [27] have studied the impact of incorporating a metal matrix into PVT/PCM systems. The results show that the use of a metal matrix-based system improves the conversion efficiency by 8.6% as compared to a system without a metal matrix. Vill et al. [28] conducted experiments to compare the performance of PVT and PVT/PCM systems and found that the thermal performance of PVT/PCM systems varies between 22 and 30%, while that of PVT systems varies between 18 and 25%. Both systems have an electrical performance between 14 and 16%. Colarossi et al. [29] studied the use of PVT/PCM systems to improve the performance of PV and hot water generated for domestic use throughout the year, but did not examine the system's performance at the specific location. They suggested that the continuous increase in panel temperature in PV/PCM systems during hot weather can be addressed by implementing a temperature regulation strategy in PVT/PCM systems. Xu et al. [30,31] conducted experiments to investigate the performance of PVT/PCM systems using thermal control techniques in outdoor conditions in Shanghai, China. They used dual-axis tracking for 30 min at 45 °C and 50 °C and found that a flow rate of 1.7 lit/min and dual-axis tracking system performed better at 45 °C. Further, an appropriate intermittent thermal control technique may significantly enhance the energy conversion performance of PVT/PCM systems. Hossain et al. [32] examined the performance of PVT/PCM systems using Lauric acid as PCM. The flow rate varied from 0.5 to 4 lit/min. The results show that the electrical efficiency obtained is 14.42% for PVT/PCM systems, which was 4.72% higher than the reference PV at a flow rate of 4 lit/min. Kazemian et al. [33] developed a 3-D numerical model to analyze the daily and monthly performance of PVT/PCM systems in Shanghai, China. They found that the percentage of melted PCM is low during the months of November, December, January, and February due to the low ambient temperature. Yang et al. [34]. Compared the performance of PVT systems with and without PCM. When the volume flow rate of water was 0.15 m^3/hr and solar intensity was 800 W/m^2, the PVT system that incorporate PCM had 14% higher energy-saving efficiency compared to the PVT system without PCM. Fayaz et al. [35] conducted a study that compare the performance of PVT and PVT/PCM systems using a serpentine flow under controlled indoor conditions. It is found that the maximum panel temperature reduces by 13.77 °C in the PVT/PCM system and 8.31 °C in the PVT system, with a flow rate varying from 0.5 to 3 lit/min and an inflow water temperature was 27 °C. Khodadadi et al. [36] conducted a numerical study on the performance of a PVT/PCM system using trapezoidal fins and nano-enhanced phase change material (NEPCM). The system was tested with varying concentrations of nanoparticles (Al_2O_3) and different numbers of fins (10, 14, 18). The results show that the system with 18 fins and a nanoparticle concentration of 0.04% by volume gives the best performance. Fu et al. [37] conducted a study on the performance of PVT systems using microencapsulated phase change material (MEPCM). The results indicate that the system using MEPCM slurry performed better than those using pure tap water or water with a PCM layer. Jurčević et al. [38] developed a new type of PVT system that uses pork fat as the PCM and evaluated its performance and cost-effectiveness in Mediterranean weather conditions. The system consisted of four cooling blocks and a container was filled with PCM, and each cooling block had a straight pipe filled with water. The study found that the system has a maximum overall efficiency of 62.2% and a cost of 0.056 to 0.083 €/kWh for the energy produced by the PVT collector.

If all technical issues have been addressed, the question remaining is whether the expenditure incurred in performance improvement of PV system is justified or not?. In fact, the biggest challenge in making a PCM-based cooling system commercially viable is determining its financial visibility. Smith et al. [39] found that using PCM cooling in PV systems is not cost-effective and that the heat stored during phase transition cannot be utilized by PV/PCM systems. Nižetić et al. [40] also reported that the high cost of PCM material makes it an unfeasible choice for passive cooling. Arici et al. [41] identified that initial material cost of PCM is barrier in PV/PCM system to be economically viable. Previous studies have shown that the use of PCM alone to cool PV panels is not cost-effective and the heat stored during phase transition cannot be utilized by PV/PCM systems [42]. However, integrating PCM into a photovoltaic thermal (PVT) system, known as PVT/PCM, has been found to be more efficient in extracting heat from PV panels and results in a

Table 1
Dimensions of PV/PCM and PVT/PCM system.

Components	Reference PV/PCM system		Proposed PVT/PCM system			
	PV panel (mm)	PCM container (mm)	PV panel (mm)	PCM container (mm)	Aluminum plate (mm)	Water container (mm)
Length (H)	120	120	120	120	120	120
Thickness (T)	5	50	5	50	1	20, 30 and 40
Width (W)	120	120	120	120	120	120

Table 2
Thermophysical properties of pot water, aluminum and RT42 [24].

Property	PCM (RT42)	Pot water	Aluminum	Unit
Solid density, ρ_s	880	–	2675	kg/m^3
Liquid density, ρ_l	760	998	–	kg/m^3
Thermal conductivity, k	0.2	0.6	211	W/m-K
Latent heat, L_h	165	–	–	kJ/kg
Dynamic viscosity, μ	0.0235	0.001003	–	kg/m-s
Heat capacity, C_p	2000	4182	903	J/kg-K
Coefficient of thermal expansion, β	0.0001	–	–	K^{-1}
Solidus temperature, T_s	311	–	–	K
Melting temperature, T_l	315		–	K
Water temperature	–	298	–	K

more stable electric and thermal energy output [42]. An investigation by Browne et al. [43] found that a PVT/PCM system could extract seven times more heat from a PV panel than a PVT system alone. Improvements in collector designs and considering economic, and environmental assessments are necessary for the implementation of PCM in PVT systems. A need for improvements in collector designs and proper utilization of PCM heat has been highlighted in the analysis by Nižetić et al. [12] to make PVT/PCM systems more cost-effective and environmentally sustainable.

In the present study, the heat stored in PCM during phase transition can be utilized through PVT/PCM systems and improvement in its performance is discussed. The system consists of two separate rectangular containers, one filled with PCM and the other with water. The containers are integrated with a PV panel on the backside. The main objectives of the study are; (i) to determine the appropriate water container thickness (WCT) for the proposed PVT/PCM system, (ii) to study the effect of WCT on the system's performance at different orientations, (iii) to compare the performance of PV/PCM and PVT/PCM systems. The developed numerical model was initially validated with experimental data and then used to analyze the effect of the optimal water container thickness on the system's parameters, such as PCM melt fraction, PV panel temperature, PCM melting behavior, and thermal energy generation at different orientations.

2. Numerical description

2.1. Geometrical model

A 2D geometric model of a reference PV/PCM and a proposed water-based PVT/PCM system at an inclination angle of β is shown in Fig. 2(a) and (b). The rectangular container filled with PCM and water is integrated with a PV panel. The PV system's different layers are considered as a single unit, and constant heat flux is applied. Table 1 lists the detailed dimensions of the reference PV system and proposed PVT/PCM system. The performance parameters are studied at various system orientations with the best suitable thickness of the water container. In the numerical study, RT42 is considered as the PCM, and the thermophysical properties of RT42, earthen pot water, and aluminum are listed in Table 2. The study uses earthen pot water, as it has a temperature of 5–7 °C lower than ordinary water [44].

2.2. Governing equations

The complexity involved in solving deforming grid technique during the phase change is overcome by Voller and Prakash [45] by easily implementing a fixed grid technique with enthalpy porosity method during phase transition from solid to liquid. The Boussinesq approximation was implemented to compute density variation as a function of temperature as given by equation (1);

$$\rho = \rho_l[1 - \beta(T - T_l)] \tag{1}$$

The momentum equation involving the relationship of buoyancy forces is given by equation (2);

$$-\rho g = \rho_l g[\beta(T - T_l) - 1] \tag{2}$$

where ρ_l, β, T_l and g are the liquid PCM density, thermal expansion coefficient, PCM melting temperature, and acceleration due to gravity, respectively.

The conservation of mass, momentum and energy equations used to solve PV/PCM and PVT/PCM system are as follows [29];

$$\frac{\partial \rho}{\partial t} + \nabla.(\rho u_i) = 0 \tag{3}$$

$$\frac{\partial(\rho u_i)}{\partial t} + \nabla.\left(\rho u_i u_j\right) = \mu \nabla^2 u_i - \nabla p + \rho g + S_i \tag{4}$$

$$\frac{\partial}{\partial t}(\rho H) + \nabla.(\rho u_i H) = \nabla.(k \nabla T) + S_h \tag{5}$$

where ρ, μ, p, H, k and $u_i and u_j$ are the density, dynamic viscosity, pressure, enthalpy, thermal conductivity, and velocity component in i and j- direction respectively, S_i and S_h are the source terms.

The total enthalpy H is the sum of sensible and latent enthalpy and is defined as follows:

$$H = h_s + \Delta H \tag{6}$$

where h_s and ΔH are the sensible and latent enthalpy which are defined as follows:

$$H = h_{ref} + \int_{T_{ref}}^{T} C_p \Delta T + \gamma L_h \tag{7}$$

where reference enthalpy is T_{ref} at the given T_{ref} and latent enthalpy changes between zero for solid and L_h for liquid as $0 < \gamma$ less than 1. γ is the melt fraction during phase transition from solid to liquid when PCM temperature is $T_s < T < T_l$ defined as follows:

$$\gamma = \begin{cases} 0 & if\ T < T_s \\ \dfrac{T - T_s}{T_l - T_s} & if\ T_s < T < T_l \\ 1 & if\ T < T_l \end{cases} \tag{8}$$

The source term S_i and S_h can be evaluated as follows:

$$S_i = A_{mush}\frac{(1-\gamma)^2}{(\gamma^3+\in)}u_i \tag{9a}$$

$$S_h = \rho L\frac{\partial\gamma}{\partial t} \tag{9b}$$

where A_{mush} represents the mush zone constant value (10^4 to 10^7). In the present analysis, mush zone constant considered is 10^5. To avoid division by zero $\in$ is taken as 0.001. γ signifies the phase change from solid to liquid when PCM temperature is $T_s < T < T$.

The thermal energy (Q_{th}) generated in PVT/PCM system is given by Yuan et al. [47]:

$$Q_{th} = m_w C_w \Delta T_w \tag{10}$$

where m_w, C_w and ΔT_w are the mass of water, the specific heat of water, and temperature difference between water in the container, respectively.

The solar PV panel electrical efficiency is measured in terms of cell temperature and it is calculated using the relation as follows [21]:

$$\eta_{pv} = \eta_{ref}[1 - \beta_{ref}(T_{pv} - T_{ref})] \tag{11}$$

Where η_{ref} and β_{ref} are reference efficiency and temperature coefficient at reference temperature T_{ref} respectively and η_{pv} is the photovoltaic panel efficiency at a photovoltaic temperature T_{pv}.

2.3. Boundary conditions

The incident solar energy, I_T is absorbed by a PV panel and then transferred to a PCM. The heat from the molten PCM is then transferred to water through an aluminum plate, before being released into the atmosphere. The model used in the simulation is the same as the one used in experimental work by Kamkari et al. [48]. In the numerical simulation, the following initial and boundary conditions are applied:

$$\text{At } t = 0,\ T(0) = Ti = 298K;\ \vec{u}(0) = \vec{u}_i = 0$$

i) The PV panel is exposed to a constant solar energy input of 1200 W/m^2:

$$T(t)_x = 0 - (55 + WCT)mm = T_p$$

y = 0–120 mm.

ii) Except back surface, all other sides of the PCM container, aluminum plate and water container are considered to be adiabatic:

At the top sides

$$-k\left(\frac{\delta T}{\delta y}\right)x = 0 - (55 + WCT)\,mm = 0$$

y = 120 mm.

At the bottom sides

$$-k\left(\frac{\delta T}{\delta y}\right)x = 0 - (55 + WCT)\,mm = 0$$

y = 0 mm.

The current research does not consider the variations in water density. In PV/PCM and PVT/PCM systems, the convection phenomena on the PCM and water container backside are taken into account for the numerical study. The convection coefficient varies in the range of 5–10 W/m^2. The backside of the PCM and water container was exposed to a convection coefficient of 10 W/m^2. This convection helps to improve heat transfer from the inside to the outside of the system and reduces the risk of heat-trapping once the PCM has completely melted.

2.4. Numerical procedure and validation

The numerical analysis of the system is carried out using the ANSYS FLUENT 19.2 software. The finite volume scheme is adopted to solve the conservation equations, and the ICEM package is used to create the geometric model and mesh. Fig. 3(a) and (b) show the meshes generated for the reference PV/PCM and PVT/PCM systems, respectively. The phase change process is modeled using the melting and solidification model, and the pressure–velocity coupling and pressure correction equation are solved using the SIMPLEC algorithm and the PRESTO scheme. To discretize energy and momentum equations, the second-order upwind method is used. A universal constant absolute convergence criteria of 10^{-6} is chosen for solving the governing equations. The under relaxation factors 1, 0.3, 0.7, and 1 are selected for density, pressure, momentum and energy, respectively.

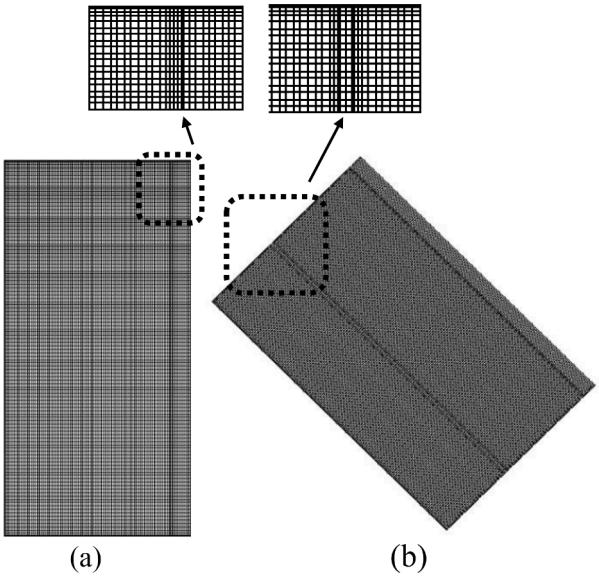


Fig. 3. Mesh generated for (a) Reference PV/PCM system; (b) PVT/PCM system.

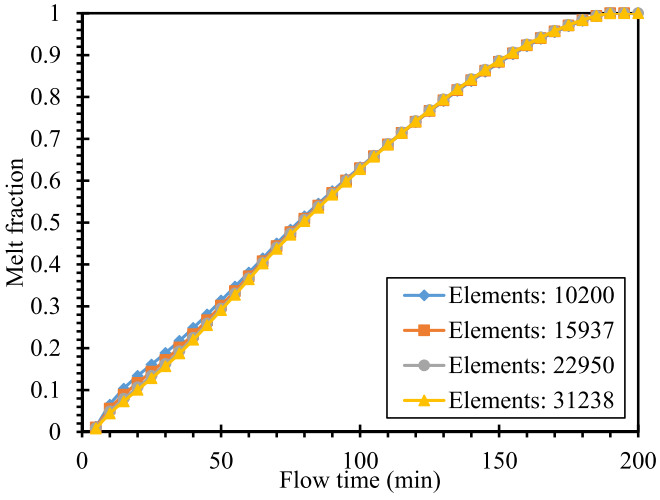


Fig. 4. Mesh independence study with a time step size of 0.01 s.

To ensure the accuracy of the computational results, a mesh and time independence analysis was performed to eliminate any effects of the number of mesh sizes. The model is tested using different mesh elements (10200, 15937, 22950, and 31238), and the results with a time step of 0.01 s are shown in Fig. 4. It is found that the maximum variation between the numerical and experimental data for the melt fraction of PCM is less than 1%. Therefore, a mesh with 22,950 elements and a time step size of 0.01 s is chosen for further numerical analysis.

To validate the developed numerical model, the results of

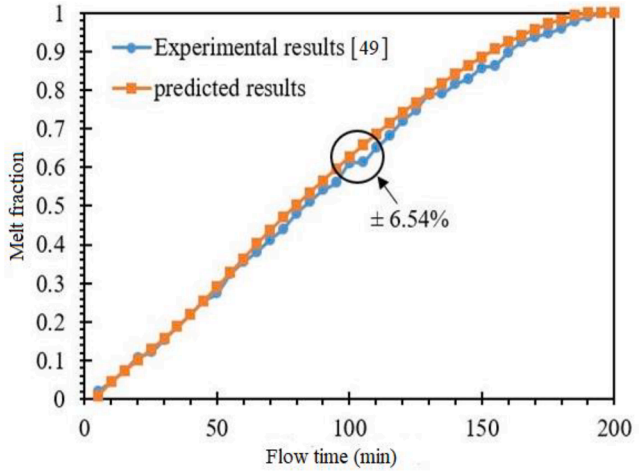


Fig. 5. PCM melt fraction comparison of experimental and numerical results.

experiments conducted by Kamkari et al. [48] are used. They studied the thermal characteristics of the melting process of PCM in a rectangular container by varying the inclination angle and wall temperature. The experiments were conducted at constant wall temperatures of 55 °C, 60 °C, and 70 °C, with system orientations of 90°, 45° and 0°. The results of the melt fraction of PCM, the temperature variation inside the container, and the heat transfer rate are obtained. The authors reported a maximum uncertainty of ± 2.9 % for the melt fraction of PCM at 90° system orientation. To validate the model, a constant wall temperature of 70 °C was applied on one side and a 90° inclination angle was considered to analyze the melt fraction of PCM in a rectangular enclosure. The numerical results of the melt fraction of PCM were found to be in good agreement with the experimental results, with a maximum error of 6.54% for the complete melting of PCM at 90° system orientation is shown in Fig. 5. This error may be due to the radiation heat transfer from the plate, which is not considered in the current numerical simulations. Furthermore, a comparison of the melting fraction contours for the experimental and numerical results is shown in Fig. 6.

3. Results and discussion

Initially, to determine the optimal thickness of the water container for the proposed PVT/PCM systems, the constant PCM thickness of 50 mm was considered and various thicknesses of the water container were investigated at a system orientation of 90°. The variations in PCM melting fraction and PV panel temperature with time are shown in Fig. 7 and Fig. 8, respectively. From these figures, it can be seen that the increase in melt fraction time and the reduction in PV panel temperature are the best suited for the PVT/PCM system when the water container thickness is 30 mm. The difference between the increase in melt fraction time and the reduction in PV panel temperature for the 30 mm and 40 mm water container thickness is minimal, around 5 min and 1 °C, respectively. However, the difference between the 20 mm and 30 mm water container thickness is more significant. Increasing the water container thickness beyond 30 mm does not produce any significant results, as the PCM reaches complete melting. Therefore, a water container thickness of 30 mm is chosen for further analysis of PVT/PCM systems at different system orientations.

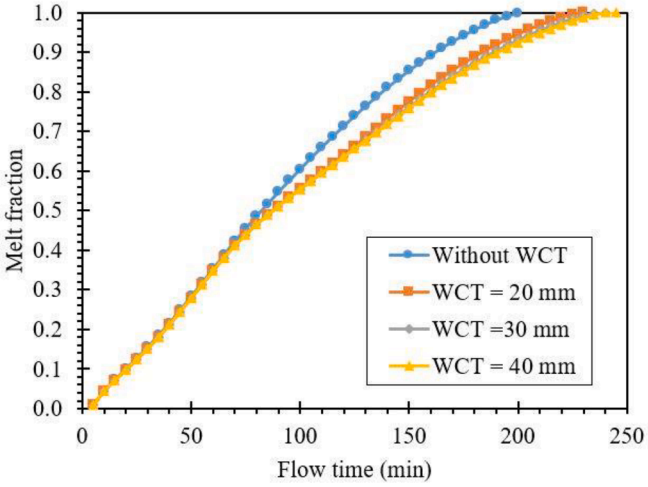


Fig. 7. Variation in PCM melt fraction with flow time for with and without WCT.

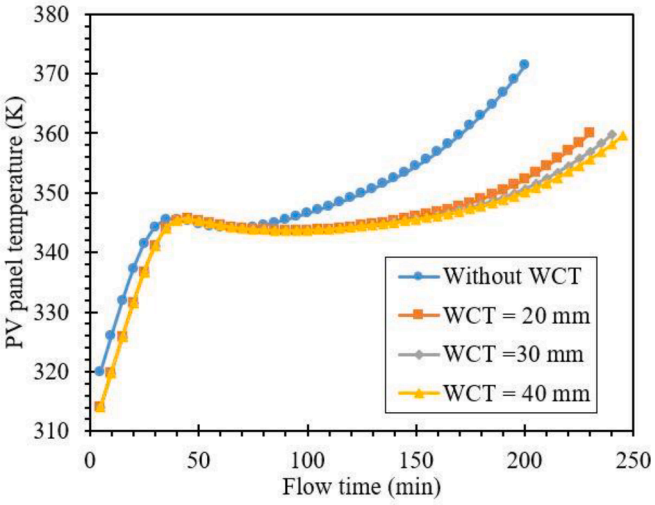


Fig. 8. Variation in average PV panel temperature with flow time for with and without WCT.

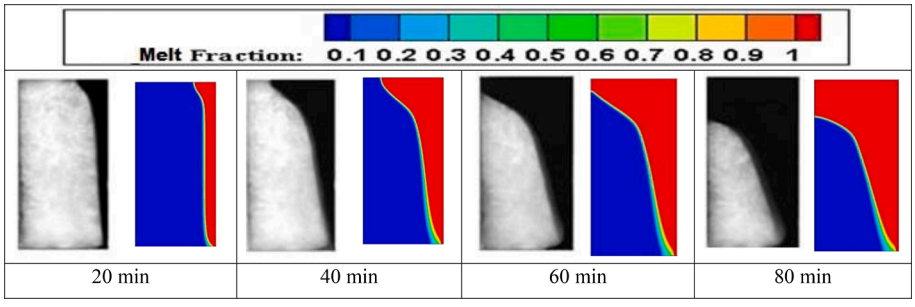


Fig. 6. Validation of numerically obtained melt fraction contours with experimental data by Kamkari et al. [48].

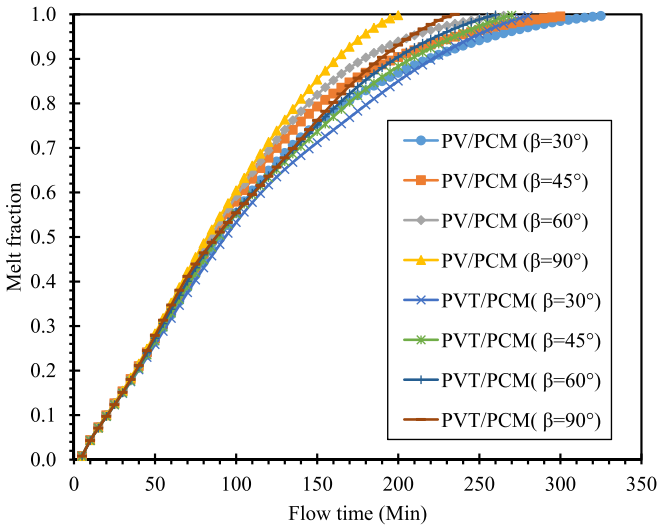


Fig. 9. Variations in PCM melt fraction with flow time at different system orientations.

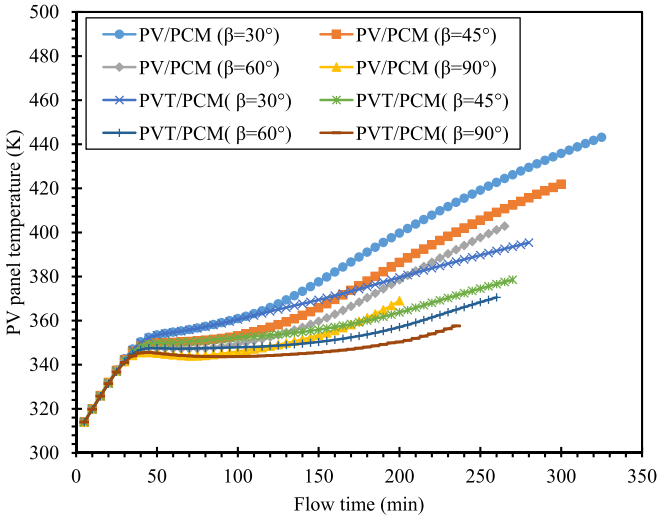


Fig. 10. Variations in average PV panel temperature with flow time at different system orientations.

3.1. Performance of PV/PCM and PVT/PCM systems

Based on the proposed methodology, the performance of PV/PCM and PVT/PCM systems has been investigated for a constant heat flux of 1200 W/m^2. In the present analysis, the effect of wind speed is not considered. The performance of both PV/PCM and PVT/PCM systems was evaluated for a PCM thickness of 50 mm and a water container thickness of 30 mm at various system orientations.

3.1.1. Melt fraction of PCM

The variations in the melt fraction of PCM with flow time at different system orientations are shown in Fig. 9. It can be seen that in both systems the total melting time of PCM decreases as the system orientation increases from 30° to 90°. This is due to a decrease in a natural convection current. Initially, heat is transferred from the PV panel to the solid PCM by conduction. The viscous forces first dominate the buoyant forces, and the liquid layer is formed parallel to the PV surface. When a particular level of melt percentage is reached, buoyant forces overcome viscous forces, and heat is transferred from the PV panel to the solid PCM by natural convection. The time taken for complete melting of PCM is

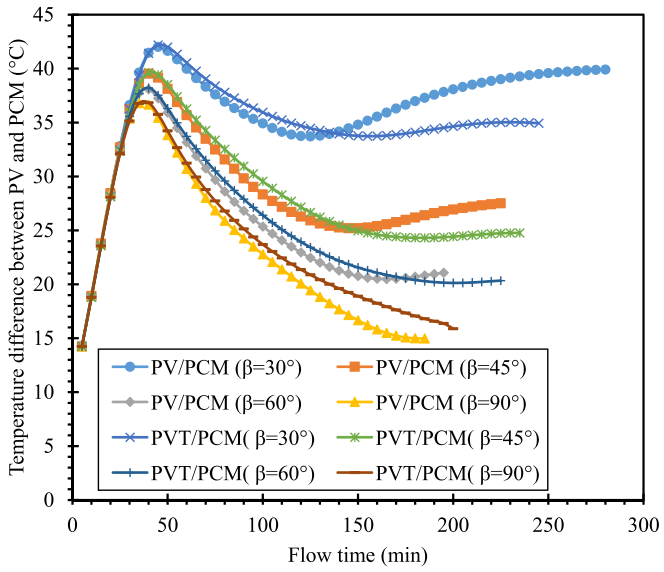


Fig. 11. Variations in the temperature difference between PV and PCM with flow time at different system orientations.

shorter in the PV/PCM and PVT/PCM systems with a 90° system orientation (200 min and 230 min, respectively) compared to a 30° system orientation (325 min and 280 min, respectively). This is due to the larger buoyant forces and increased natural convection heat transfer at a 90° orientation. The use of water storage in the PVT/PCM system increases the melting time by around 35 min at 90° system orientation and decreases melting time by around 45 min at 30° system orientation. The reason for this is that at 90° system orientation, the heat dissipation from PV to PCM, PCM to water and water to surrounding is higher due to a higher convection heat transfer. However, at 30° system orientation, the heat dissipation is less due to the reduced convection heat transfer which results in a reduction in the heat absorption from molten PCM. As a result, the total PCM melting time decreases at 30° in PVT/PCM system compared to PV/PCM system.

3.1.2. PV panel temperature

The variations in the average photovoltaic panel temperature with flow time for complete PCM melting at different system orientations are shown in Fig. 10. From the figure, it can be seen that, initially, the PV panel temperature rapidly increases due to high heat flux. However, when the PCM begins to receive PV heat, the PV temperature remains stable for a certain period of time. After 75–80 min, the heat is transferred from molten PCM to water by conduction through an aluminum plate. It was observed that in PV/PCM system, the continuous increase in the temperature of the molten PCM is transferred to water in the PVT/PCM system, resulting in a significant decrease in PV panel temperature. As a result, the average PV panel temperature is always less in PVT/PCM systems than in PV/PCM systems. Also, it was found that as the system orientation decreases from 90° to 30°, the PV cell temperature increases, due to a reduction in the natural convection effect. As the system orientation increases from 30° to 90°, the temperature difference in PV panels in both systems decreases. This is because, the PCM can absorb panel heat for a longer period of time at lower system orientations, leading to an increase in cell temperature and reduced natural convective heat transfer. The heat absorbed by PCM from panel start transferring to water after 100 min, which significantly decreases the panel temperature until complete melting of the PCM. For complete melting of PCM, the average PV temperature reduction of 23 °C and 3 °C are obtained at 30° and 90° system orientations, respectively. This indicates that the optimum inclination angle for the system can be adjusted depending on the season and location to achieve maximum temperature reduction and efficiency. The optimum inclination angle for solar

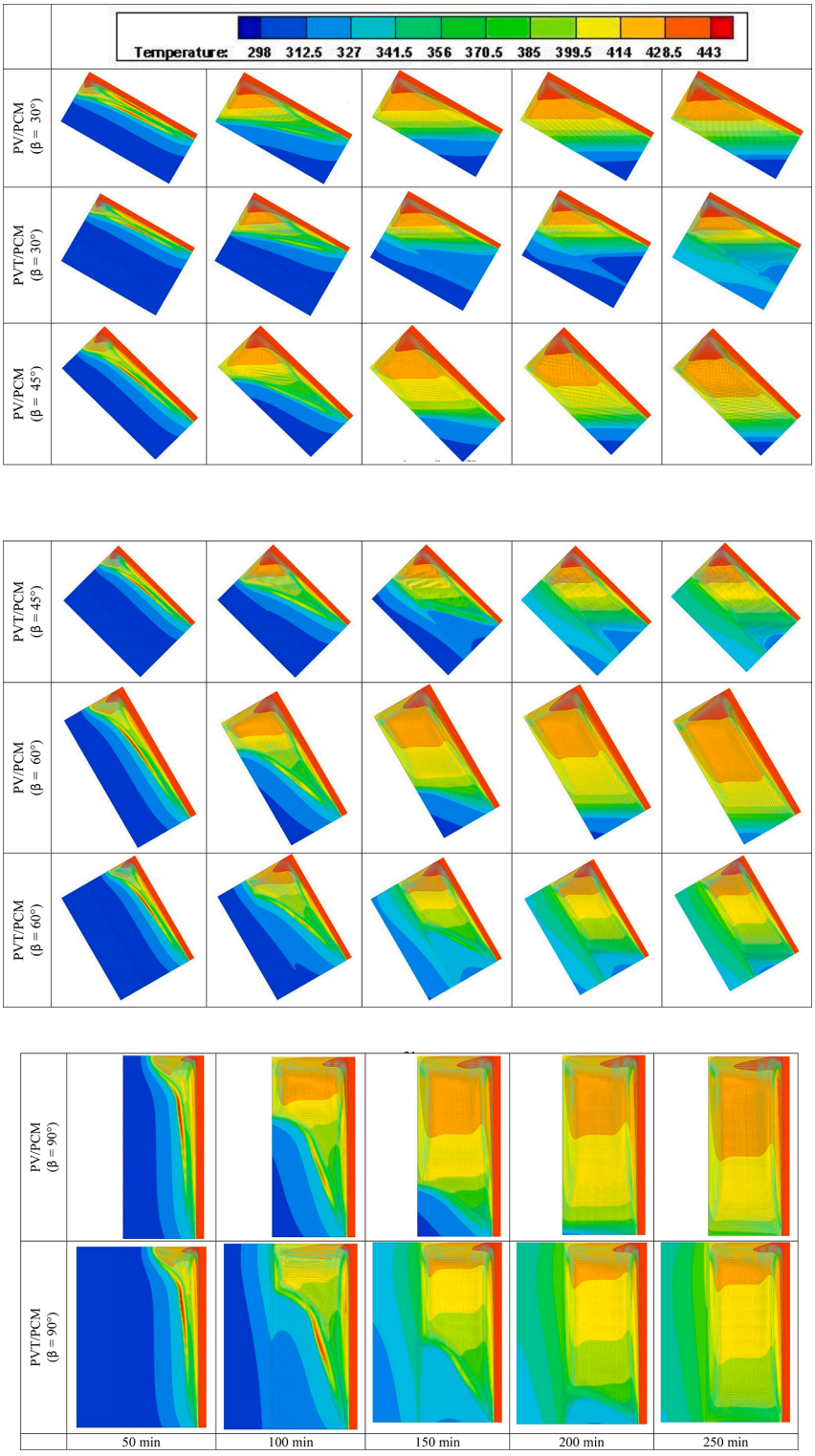


Fig. 12. Temperature isotherms for PV/PCM and PVT/PCM systems at different flow times and system orientations.

photovoltaic panels was determined using solar radiation data [49] of the location. The inclination angle depends on the season. The inclination angle is higher during the winter and lower during the summer. The angle reaches a maximum in December around 50-60° depending on the location's latitude. In spring, the best angle is 45°, while in summer it is 20°.

3.1.3. Temperature difference between PV and PCM

The variations in the temperature difference between PV and PCM with flow time at different system orientations are shown in Fig. 11. It can be seen that in both systems, the temperature difference at a lower system orientation is higher compared to a higher system orientation. This is due to the fact that at a lower system orientation, it takes a longer time (heat extraction rate is low) for complete PCM melting compared to a higher system orientation. Also, as the system orientation decreases

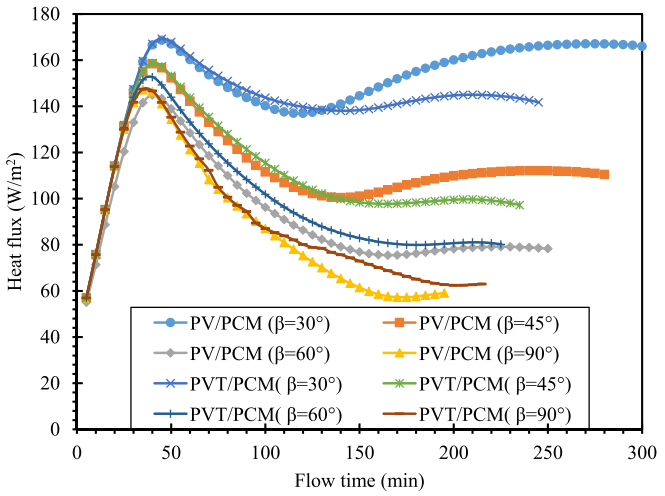


Fig. 13. Variations in heat flux with time for PV/PCM and PVT/PCM systems at different system orientations.

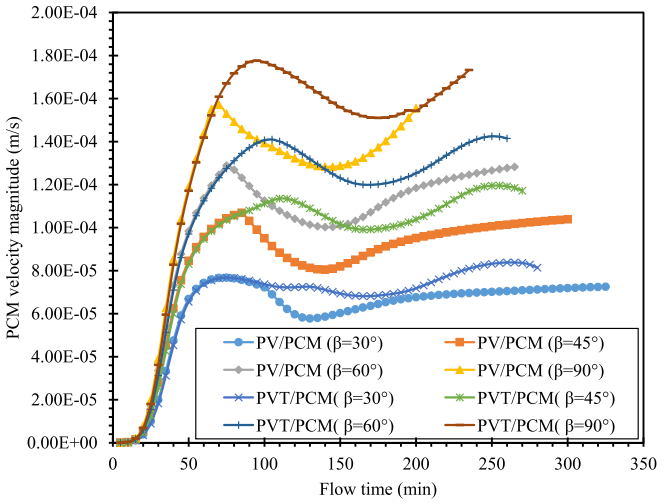


Fig. 14. Variations in average PCM velocity magnitude with flow time at different system orientations.

from 90° to 30°, the average temperature difference between PV and PCM increases. The maximum temperature difference of around 43 °C and 12 °C has been observed at 30° and 90° system orientations, respectively.

The heat propagation from the PV panel to the PCM and then from molten PCM to the water through the aluminum plate at different flow times and system orientations is shown in Fig. 12. The contour displays the solid–liquid interface and temperature distribution of the melted PCM, and the heat flow from the molten PCM to the water inside the container. Initially, a temperature profile parallel to the PV surface is observed. The upper half of the enclosure has a stronger convection current, causing the temperature on the upper side to rise faster than the bottom side temperature. This results in a faster melting rate and shorter time for complete PCM melting at a higher inclination of the system. The PCM melting starts in the upper right corner and propagates downward as the heat from the molten PCM is transferred to the water.

3.1.4. Heat flux

The variations in heat flux with flow time for the PV/PCM and PVT/PCM systems at different system orientations are shown in Fig. 13. It can be seen that the heat exchange rate between the PV panel and PCM is low at the beginning but increases as the PCM starts to melt, reaching a maximum heat flux of 168 W/m^2 at a 30° system orientation. The heat flux is higher at a lower system orientation compared to a higher system orientation. Heat is initially transferred from the PV panel to the PCM by conduction. After 45–50 min, heat is stored in the PCM as latent heat during the melting process, resulting in a decrease in heat flux for both PV/PCM and PVT/PCM systems. In the PVT/PCM system, the heat exchange rate remains stable for all system orientations during the melting of PCM, while in the PV/PCM system, it increases with the increase in PV temperature. This is because the PV/PCM system experiences a continuous increase in the temperature of the molten PCM.

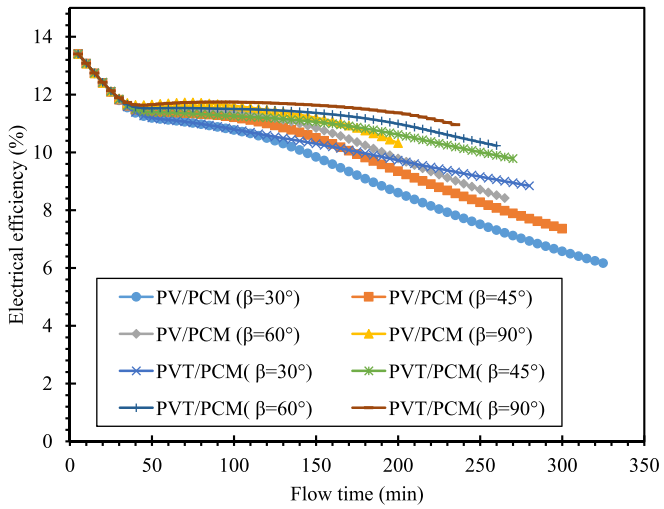


Fig. 15. Variations in PV panel electrical efficiency with flow time at different system orientations.

3.1.5. PCM velocity magnitude

The variations in the velocity of the molten PCM with flow time at different system orientations are shown in Fig. 14. Both PV/PCM and PVT/PCM systems show that the velocity of the molten PCM is higher at higher system orientations. This is due to the increased fluid flow (Natural convection) during the PCM melting process at higher orientations. The PVT/PCM system has a higher velocity of the molten PCM than the PV/PCM system because the water in the PVT/PCM system removes heat more quickly from the molten PCM, which increases the fluid flow and thus the velocity of the molten PCM. The velocity of the molten PCM decreases as the system orientation decreases from 90° to 30°. For PVT/PCM system, the maximum velocity of 0.000173 m/sec and 0.0000814 m/sec is observed at 90° and 30° system orientations, respectively.

3.1.6. Electrical efficiency of PV panel

The variations in the electrical efficiency of the PV panel with flow time at different system orientations are shown in Fig. 15. The electrical efficiency is determined in terms of cell temperature (using equation (11) and is affected by the temperature of PV cell surpassing the working temperature. The electrical efficiency increases as the system orientation increases from 30° to 90° for both PV/PCM and PVT/PCM systems. The higher gain in electrical efficiency is seen at 30° system orientation, followed by 45°, 60° and 90° system orientations. The average electrical efficiency is found to be around 9.33% and 10.43% for PV/PCM and PVT/PCM systems at 30° system orientation, respectively, while at 90° system orientation, the average efficiencies obtained are 11.52 % and 11.69 % for PV/PCM and PVT/PCM systems, respectively. Compared to PV/PCM system, the PVT/PCM system shows an increase in electrical efficiency by 11.79 % and 1.48 % at 30° and 90° system orientations, respectively. However, the average decrease in electrical efficiency at 90° system orientation is due to the shorter total PCM melting time.

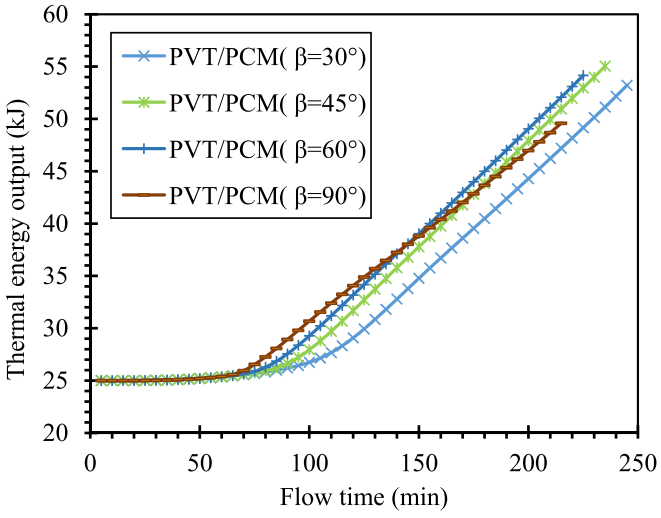


Fig. 16. Variations in thermal energy output with flow time at different system orientations.

3.1.7. Thermal energy output

The variations in thermal energy output with flow time at different system orientations are shown in Fig. 16. When water is used to extract heat from the melted PCM, the temperature of the water in the container increases, producing thermal energy. The highest thermal energy output is obtained when the system is oriented at an angle of 45°. This results in a rapid increase in water temperature. There is also a slight difference in thermal energy production when the system is oriented at 30° versus 45° due to differences in thermal stratification at the end of the melting process, which affects the total amount of sensible heat stored. This energy produced can be used for other applications. Overall, the maximum thermal energy output is obtained when the system is tilted at 45°.

4. Conclusions

This numerical study demonstrates the importance of using a PVT/PCM system over a PV/PCM system in hot environmental conditions. The system performance is evaluated using a PCM thickness of 50 mm and a WCT of 30 mm at various orientations. Key parameters such as PCM melt fraction, average PV panel temperature, PCM melting behavior and temperature distribution, molten PCM velocity, PV panel electrical efficiency, and thermal energy output were analyzed. The important findings are as follows:

- The proposed PVT/PCM system demonstrates the best performance with a WCT of 30 mm. Increasing the WCT beyond 30 mm does not lead to a significant improvement in performance.
- The melting rate of PCM is higher at higher system orientations and the time required for complete PCM melting increases as the system orientation decreases from 90° to 30°.
- As the system orientation increases from 30° to 90°, the PV panel temperature decreases. The maximum average PV panel temperature decreased by 23 °C and 5 °C at 30° and 90° system orientations, respectively, compared to PV/PCM system for complete PCM melting.
- The difference in PV and PCM temperature increases as the system orientation decreases from 90° to 30° and is higher at lower system orientations.
- The molten PCM velocity magnitude inside the container is always higher at 90° system orientation and decreases as the system orientation decreases from 90° to 30°.
- The PVT/PCM system shows a maximum electrical efficiency improvement of 3.57% at 30° system orientation, followed by 45°, 60° and 90°, compared to PV/PCM system for complete PCM melting.
- The water effectively carries heat from the molten PCM, which improves the system's performance and produces hot water that can be used for other applications.
- The proposed system is effective in controlling PV panel temperature and improving performance in hot climatic conditions.
- It is recommended that after the complete melting of PCM, the water in the container should be removed and replaced with fresh cold water.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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